

Lab1 Intro



Lab1: Agenda

- Διαδικαστικά
- DBMS Installation
- Lab schema
 - Installation
 - Explanation

- Παρουσίες
- Ωράριο
- Εργασία Μαθήματος

DBMS (Data Base Management System)

1. Supported DBMS

-  <https://www.mysql.com/>

-  <https://mariadb.org/>

-  <https://www.postgresql.org/>

DBMS (Data Base Management System)

2. 💡 Un Supported DBMS:



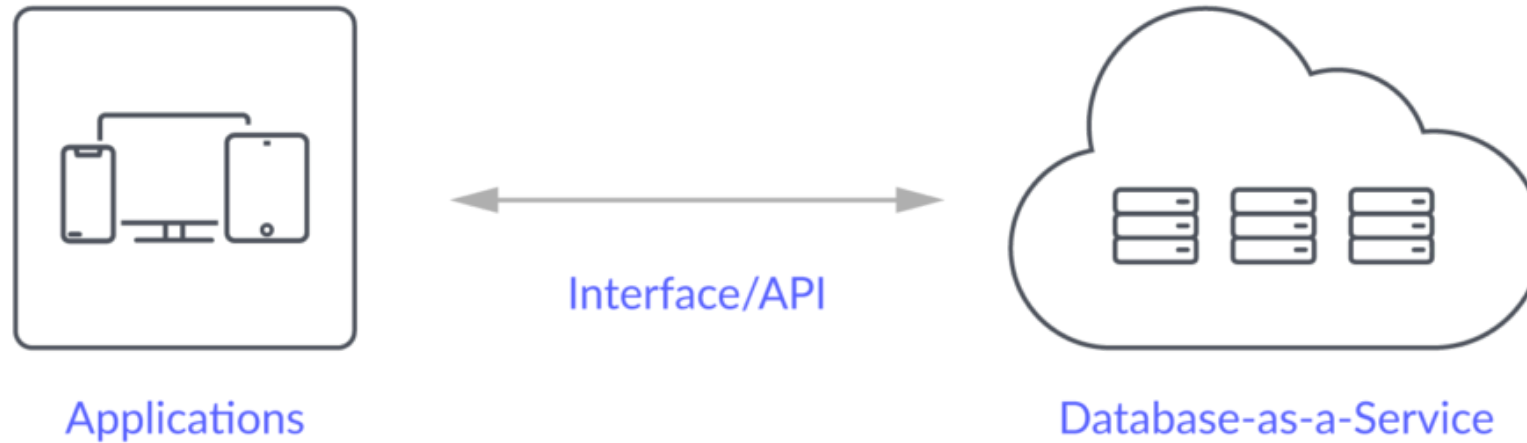
DBMS (Data Base Management System)

3. No GO

-  SQLite
- Key/value stores
- NoSQL Databases
- Column-oriented
- Hierarchical databases
- Document-oriented databases
- Graph databases
- Object-Oriented databases
- Map/reduce frameworks
- Semantic data stores
- ... any other kind

DBMS *Database as a Service*

4. 💣 are u serious?



DBMS (Data Base Management System) Choice ?

- mysql-vs-postgresql comparison 📖 read:
<https://www.digitalocean.com/community/tutorials/sqlite-vs-mysql-vs-postgresql-a-comparison-of-relational-database-management-systems>

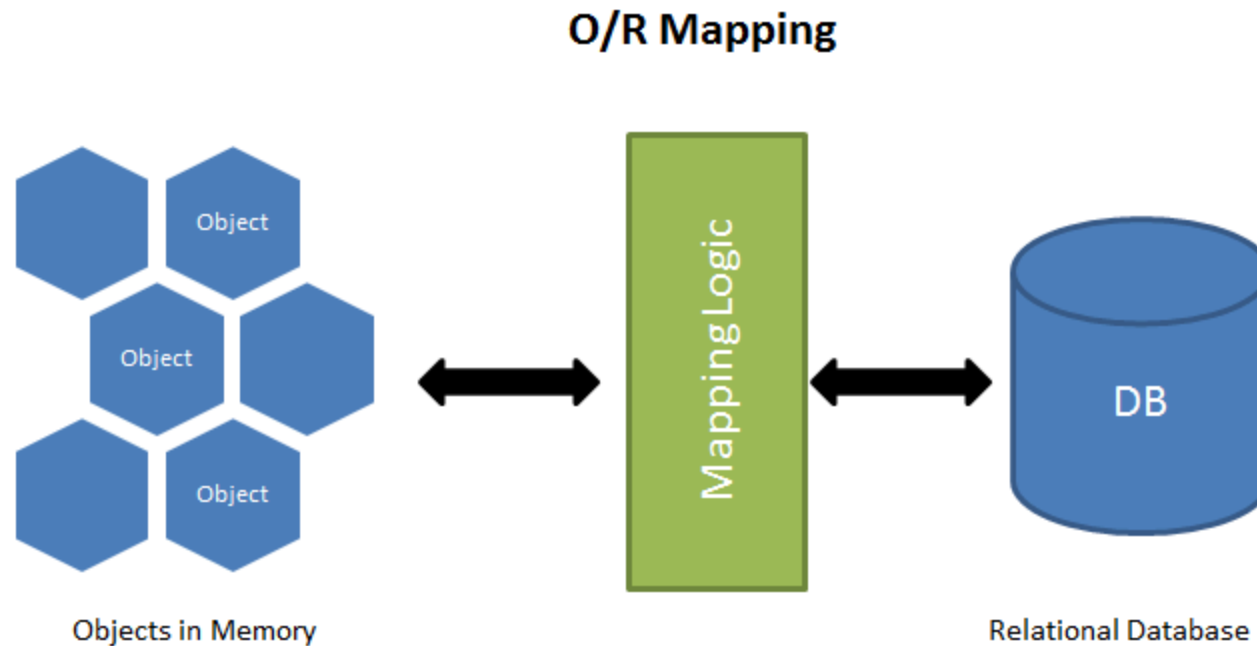
DBMS Graphical Clients

- MySQL Workbench 
- Pgadmin 
- Database Workbench 
- Dbeaver 
- HeidiSQL 
- DbVisualizer 
- Phpmyadmin 
- ... any other tool ...

LIST  <https://mariadb.com/kb/en/graphical-and-enhanced-clients/>

ORM Frameworks 🔗

Disadvantages of ORM tools generally stem from the high level of abstraction obscuring what is actually happening in the implementation code. Also, heavy reliance on ORM software has been cited as a major factor in producing poorly designed databases



Development Tools ORM CODEGEN

- If still unsure

- OK

```
var sql = "SELECT id, first_name, last_name, phone, birth_date FROM persons WHERE id = 10";  
var result = context.Persons.FromSqlRaw(sql).ToList();  
var name = result[0]["first_name"];
```

- Warning: NOT OK

```
var person = repository.GetPerson(10);  
var firstName = person.GetFirstName();
```

Development Tools - Data generators

Dummy Data για την εργασία

1. <https://generatedata.com/>
2. <https://www.mockaroo.com/>
3. <https://cobbl.io/>
4. ... add you own ...

Dev Enviroment - Web Dev Stack

- XAMP/ LAMP/ WAMP **vs** MERN / MEAN stack



- Bundles with Apache, MySql/ MariaDB, PHP

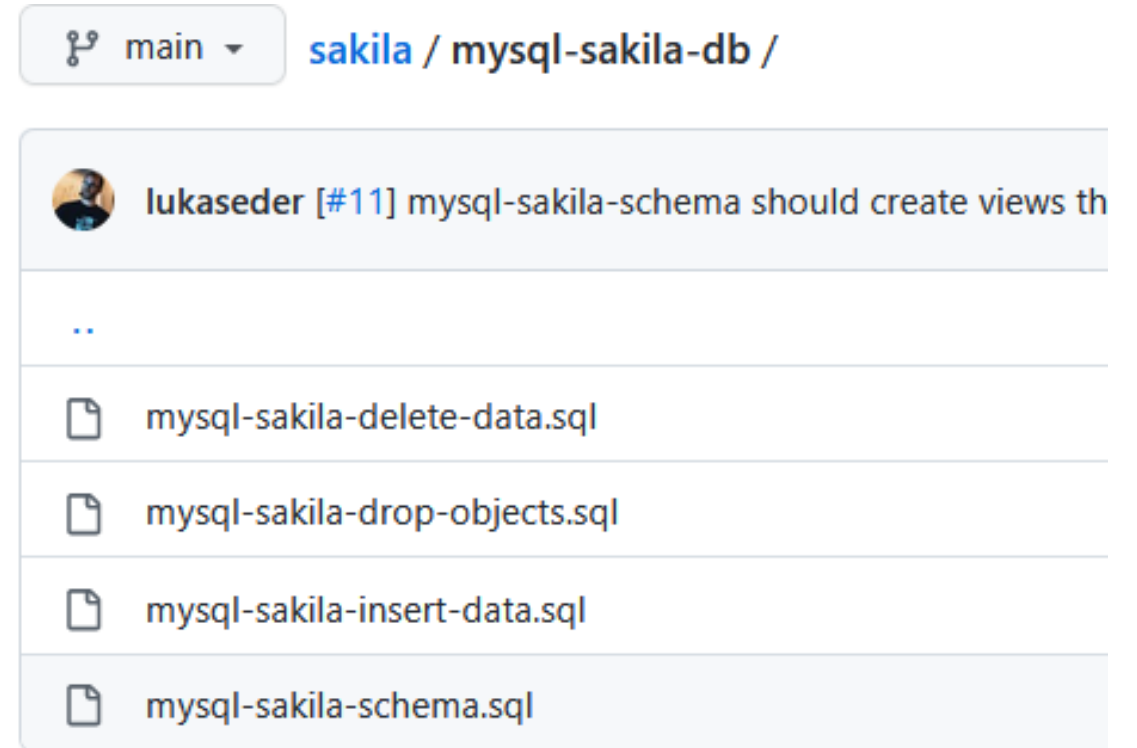
1. [Windows XAMP](#) (Also for Linux, OS X)
💡 Tutorial: [installation-process-of-xamp](#)
2. [Mac MAMP](#) (Also for Windows)
💡 Tutorial: [how-to-install-mamp-on-your-mac](#)
3. [Linux LAMP]
💡 Tutorial: [LAMP Stack for Ubuntu install](#)
4. [Docker](#)
💡 Tutorial: [LAMP Stack with Docker Compose](#)

 **git** <https://git-scm.com/>

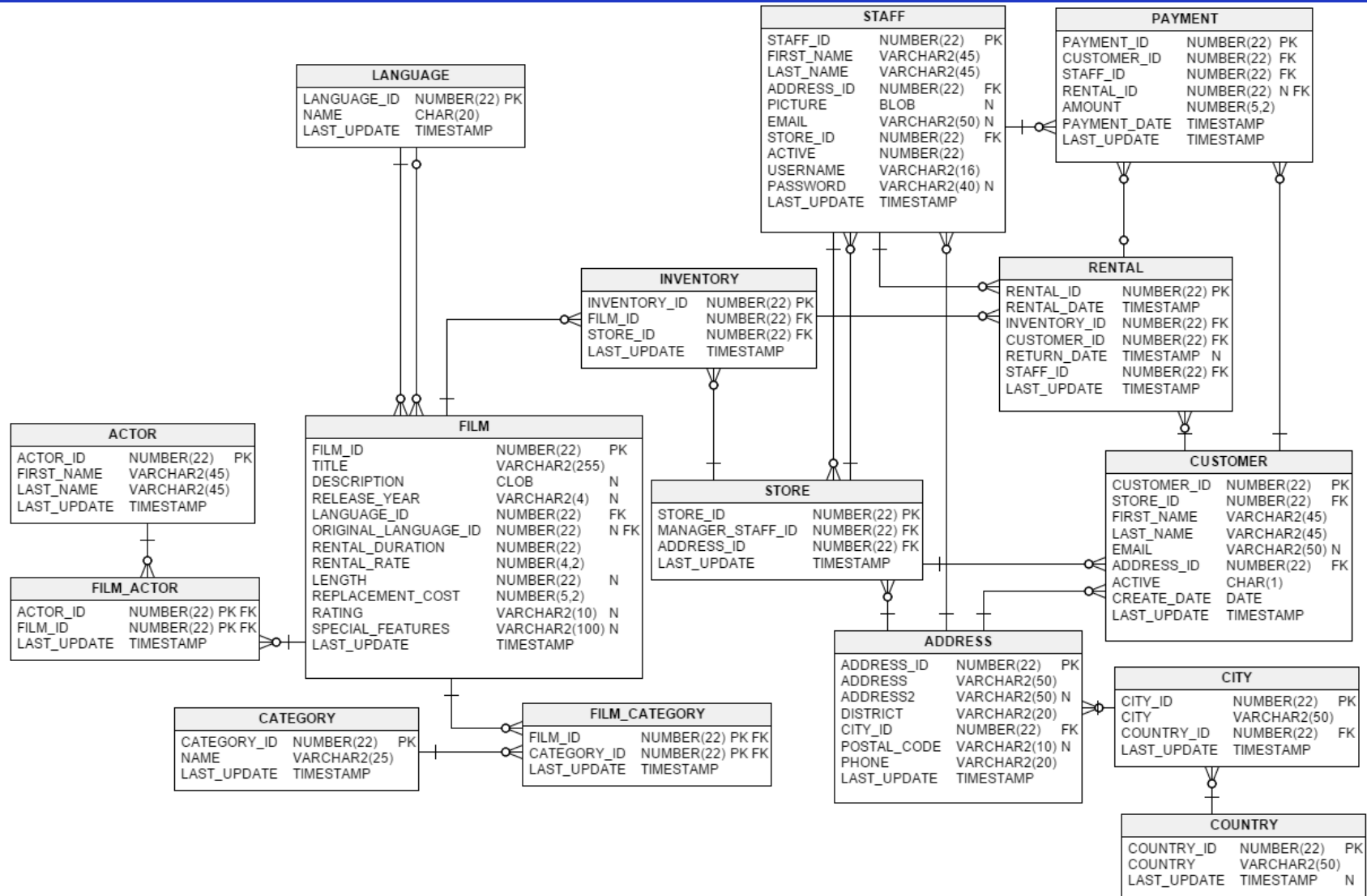
 <https://github.com/>

Dev Enviroment - sample lab schema

- The **Sakila** example database
<https://github.com/jOOQ/sakila>
- DVD Rental store 🗑️
- SQL scripts
 - i. delete data (for mysql)
 - ii. drop (for mysql)
 - iii. *create*
 - iv. *insert*



sample lab schema



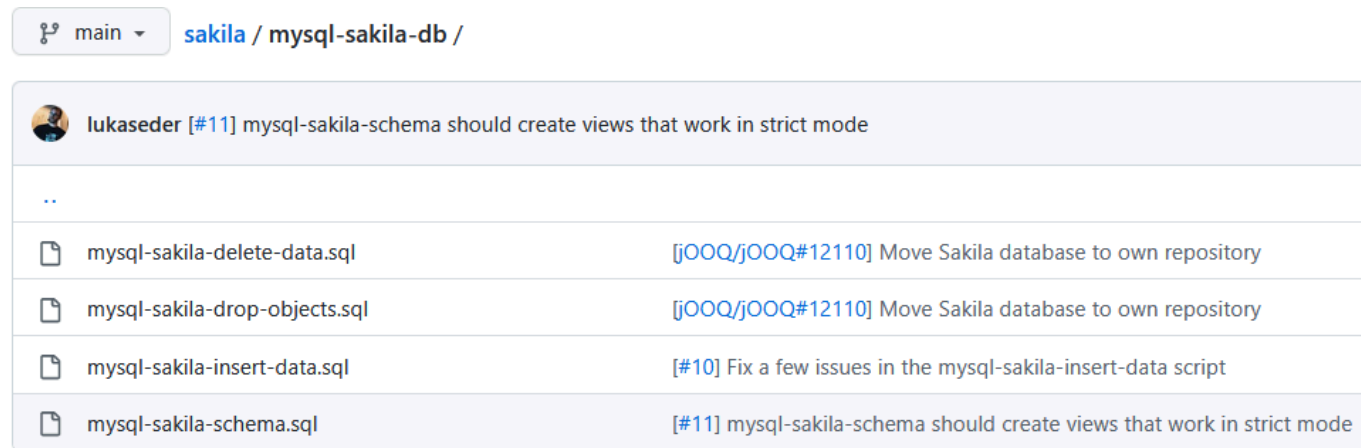
Installation Session

1. Install {W|L|M} AMP
2. Connect
3. Install Lab schema
4. Walkthrough

Installation Session - Connect & Install Lab schema

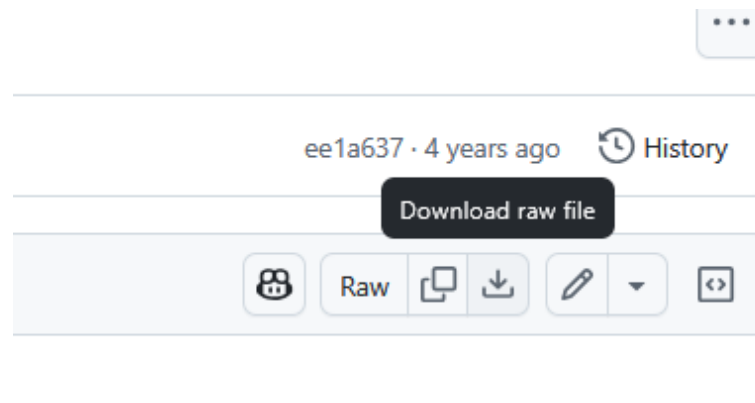
- Lab

i. Navigate to <https://github.com/jOOQ/sakila/tree/main/mysql-sakila-db>



Installation Session - Connect & Install Lab schema

- Lab 
 - 2. save files
 - mysql-sakila-schema.sql
 - mysql-sakila-insert-data.sql
- to a folder e.g. c:\temp



sakila / mysql-sakila-db / mysql-sakila-schema.sql 

 lukaseder [#11] mysql-sakila-schema should create views that work in strict mode

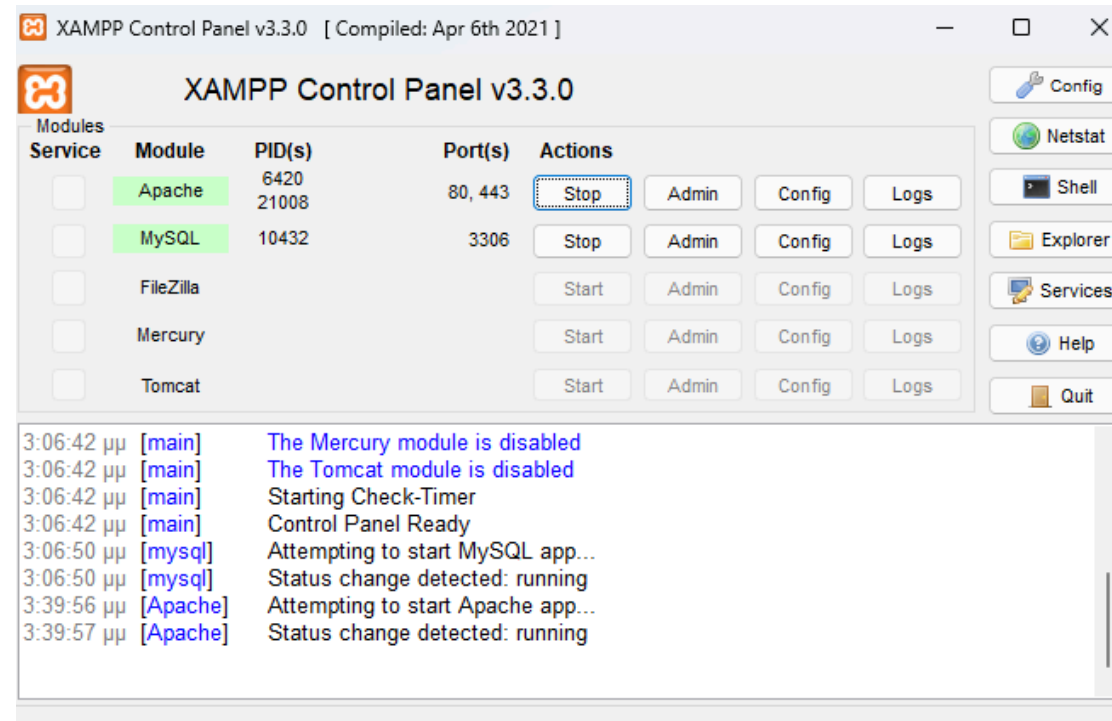
Code Blame 644 lines (535 loc) · 22.5 KB

```
1
2  -- Sakila Sample Database Schema
3  -- Version 0.8
4
5  -- Copyright (c) 2006, MySQL AB
6  -- All rights reserved.
7
8  -- Redistribution and use in source and binary forms, with or without modification, are permi
9
10 -- * Redistributions of source code must retain the above copyright notice, this list of cor
11 -- * Redistributions in binary form must reproduce the above copyright notice, this list of
12 -- * Neither the name of MySQL AB nor the names of its contributors may be used to endorse c
13
14 -- THIS SOFTWARE IS PROVIDED BY THE COPYRIGHT HOLDERS AND CONTRIBUTORS "AS IS" AND ANY EXPRES
15
16
17 SET @OLD_UNIQUE_CHECKS=@@UNIQUE_CHECKS, UNIQUE_CHECKS=0;
18 SET @OLD_FOREIGN_KEY_CHECKS=@@FOREIGN_KEY_CHECKS, FOREIGN_KEY_CHECKS=0;
19 SET @OLD_SQL_MODE=@@SQL_MODE, SQL_MODE='TRADITIONAL';
20
21 DROP SCHEMA IF EXISTS sakila;
22 CREATE SCHEMA sakila;
23 USE sakila;
24
25 --
26 -- Table structure for table `actor`
27 --
28
29 CREATE TABLE actor (
30   actor_id INT UNSIGNED NOT NULL AUTO_INCREMENT,
31   first_name VARCHAR(45) NOT NULL,
32   last_name VARCHAR(45) NOT NULL,
33   last_update TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
34   PRIMARY KEY (actor_id),
35   KEY idx_actor_last_name (last_name)
36 )ENGINE=InnoDB DEFAULT CHARSET=utf8;
37
38 --
39 -- Table structure for table `address`
40 --
```

Installation Session - Connect & Install Lab schema

- Lab 

3. Run XAMP and start "MYSQL" and "Apache"



Installation Session - Connect & Install Lab schema

- Lab

4. Open CMD prompt and navigate to "C:\Program Files\MariaDB 10.5\bin"

```
c:  
cd "C:\Program Files\MariaDB 10.5\bin"
```

- Lab

5. `mysql -u root -p`

```
C:\Program Files\MariaDB 10.6\bin>mysql -u root -p  
Enter password:  
Welcome to the MariaDB monitor.  Commands end with ; or \g.
```

 [Windows cmd prompt](#)

Installation Session - Connect & Install Lab schema

- Lab

6. run "source c:\temp\mysql-sakila-schema.sql"

```
MariaDB [(none)]> source c:\temp\mysql-sakila-schema.sql
```

- Lab

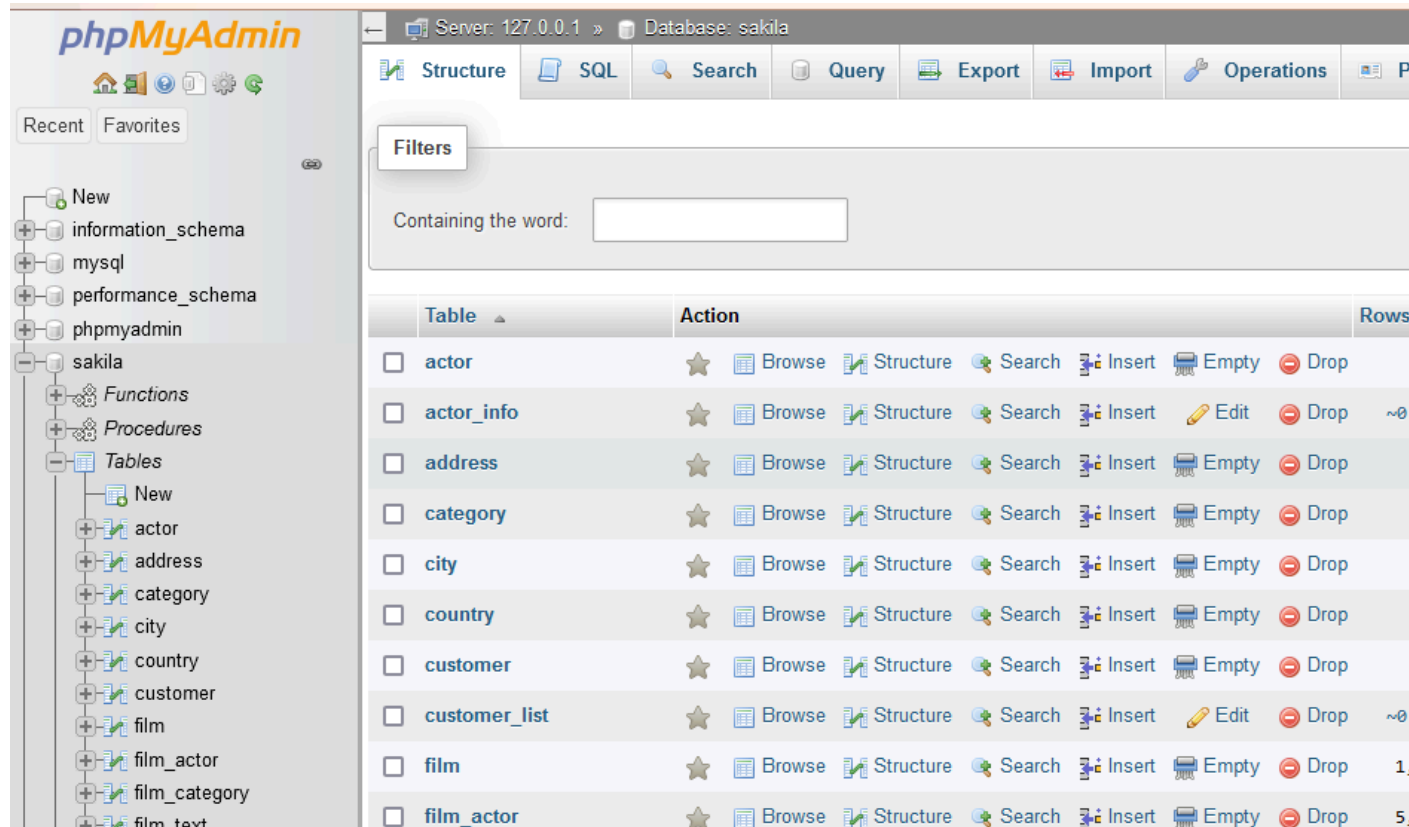
7. source c:\temp\mysql-sakila-insert-data.sql

```
MariaDB [(none)]> source c:\temp\mysql-sakila-insert-data.sql
```

Installation Session - Connect & Install Lab schema

- Lab

8. Open browser and navigate to <http://localhost/phpmyadmin/>



The screenshot shows the phpMyAdmin web interface. The left sidebar displays a tree view of databases, with 'sakila' selected under 'Recent'. The main panel shows the 'Structure' tab for the 'sakila' database. A table of tables is displayed with columns for 'Table' and 'Action'. Each table row includes a checkbox, a star icon, and links for 'Browse', 'Structure', 'Search', 'Insert', 'Empty', and 'Drop'. The tables listed are: actor, actor_info, address, category, city, country, customer, customer_list, film, film_actor, film_category, and film_text.

Table	Action	Rows
☐ actor	★ Browse Structure Search Insert Empty Drop	
☐ actor_info	★ Browse Structure Search Insert Edit Drop	~0
☐ address	★ Browse Structure Search Insert Empty Drop	
☐ category	★ Browse Structure Search Insert Empty Drop	
☐ city	★ Browse Structure Search Insert Empty Drop	
☐ country	★ Browse Structure Search Insert Empty Drop	
☐ customer	★ Browse Structure Search Insert Empty Drop	
☐ customer_list	★ Browse Structure Search Insert Edit Drop	~0
☐ film	★ Browse Structure Search Insert Empty Drop	1
☐ film_actor	★ Browse Structure Search Insert Empty Drop	5

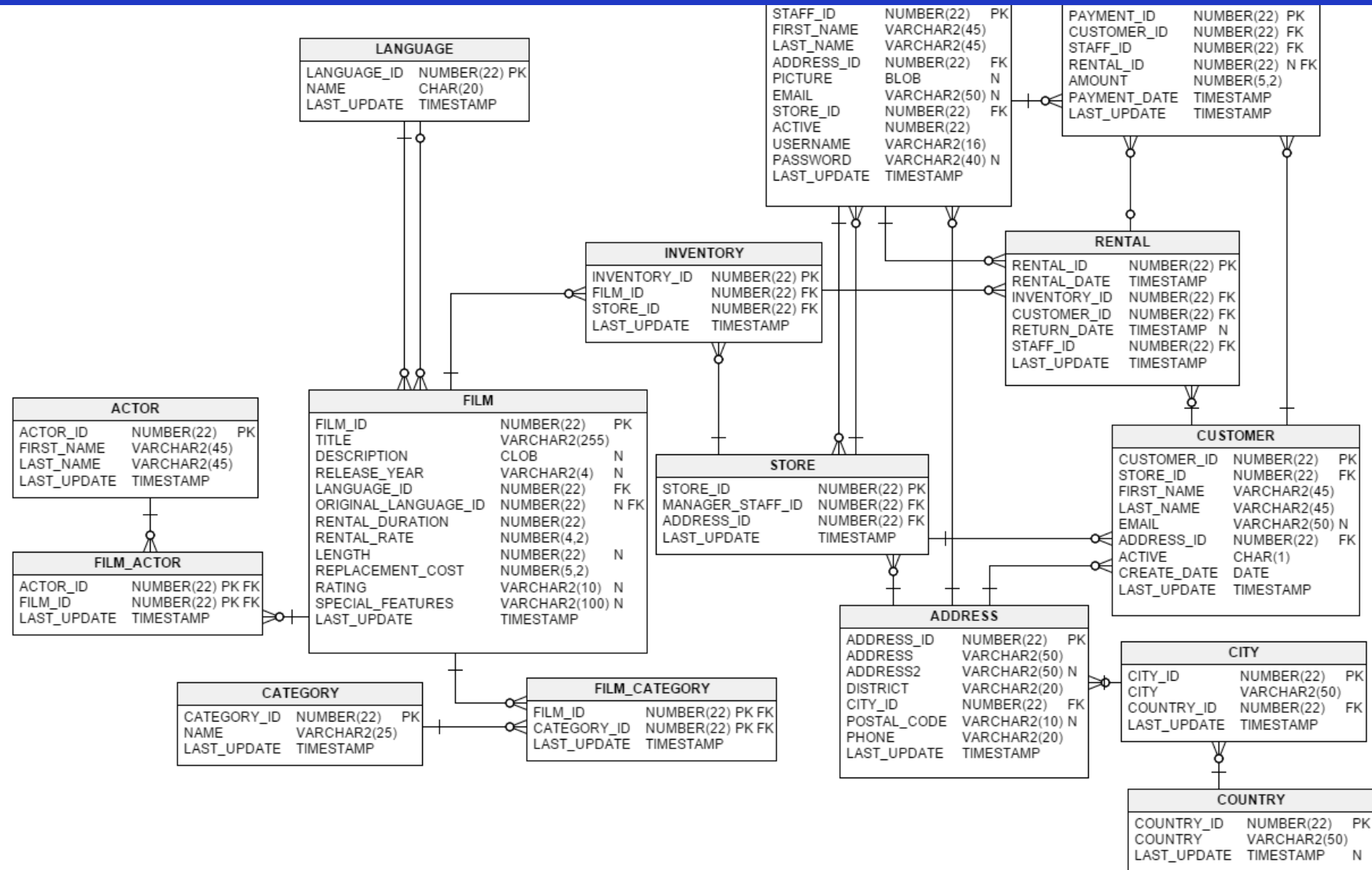
Installation Session - Connect & Install Lab schema

- Your PC 
 - Same steps as above but in **Step 3** replace install path with yours e.g.
"D:\dev_tools\xampp\mysql\bin"

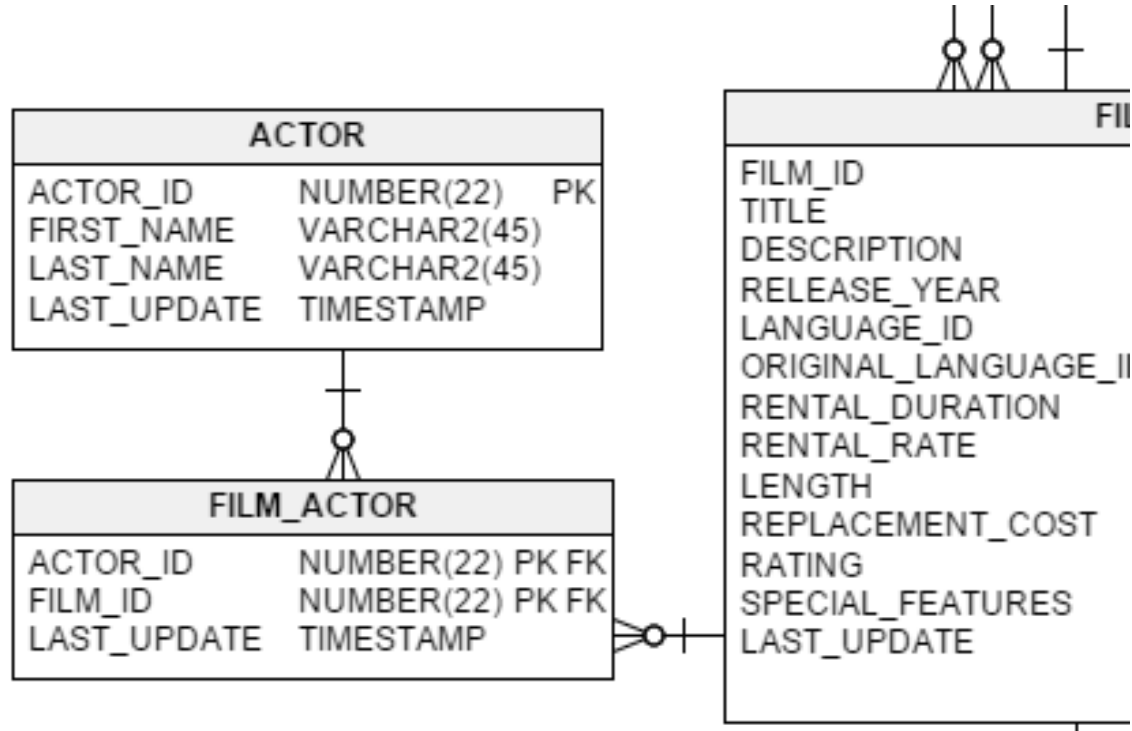
The Sakila Database: Tables 🙄

Table Name	Definition
film	A movie that has been released and can be rented
actor	A person who acts in films
customer	A person who watches films
category	A genre of films
payment	A rental of a film by a customer
language	A language spoken by the actors of a film
film_actor	An actor in a film
inventory	A film available for rental

Lab schema Explanation

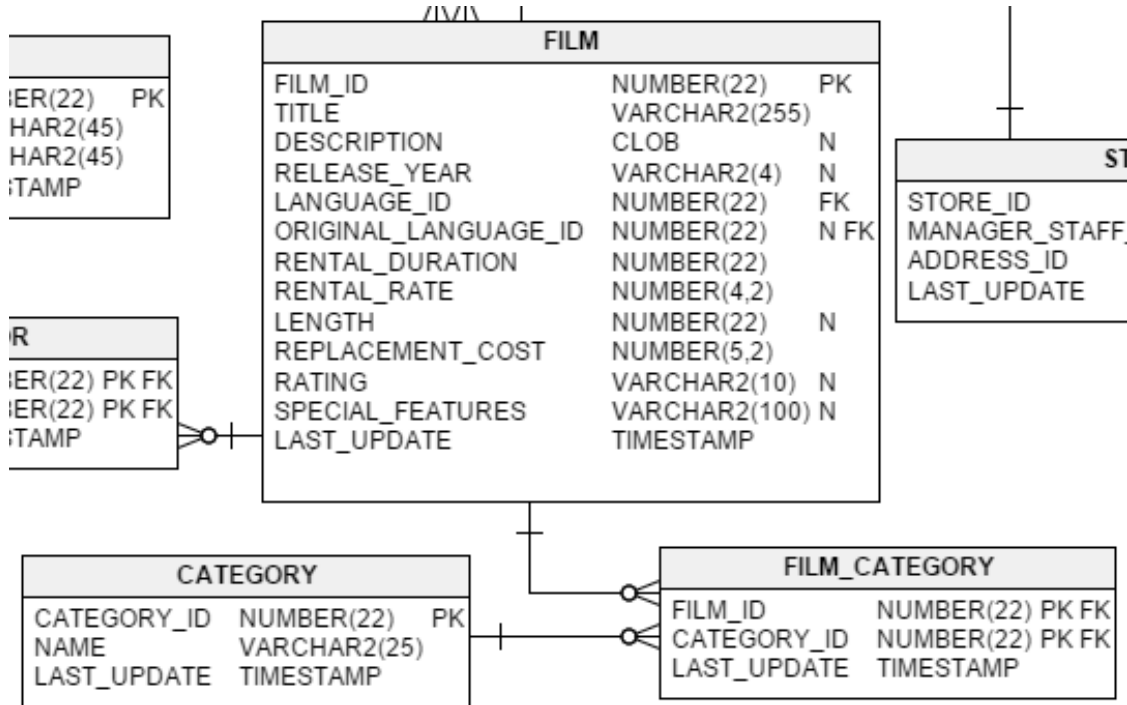


Lab schema: *Actors*



- Ο πίνακας **actor** περιέχει πληροφορίες για τους ηθοποιούς και συνενώνεται με τον πίνακα **film** μέσω του **film_actor**.
 - **actor_id** (πρωτεύον κλειδί)
 - **first_name**
 - **last_name**
- Ο πίνακας **film_actor** υποστηρίζει τις σχέσεις μεταξύ ηθοποιών και ταινιών.
 - **actor_id**
 - **film_id**

Lab schema: *Category*



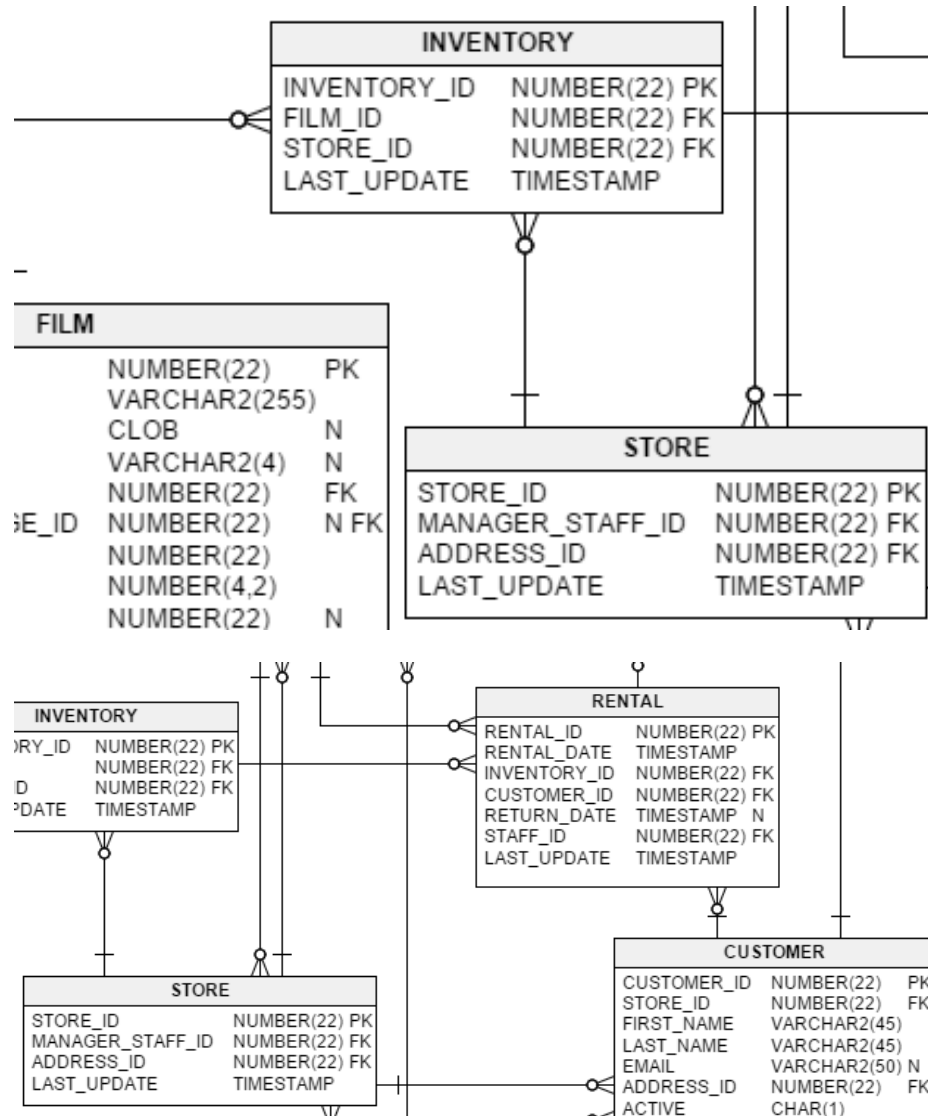
- Ο πίνακας **category** περιέχει τις κατηγορίες στις οποίες ανήκουν οι ταινίες.
 - category_id (πρωτεύον κλειδί – ενώνεται με τον πίνακα film μέσω του film_category)
 - name
- Ο πίνακας **film_category** υποστηρίζει τις σχέσεις μεταξύ ταινιών και κατηγοριών ταινιών. Ο πίνακας αναφέρεται στις ταινίες και τις κατηγορίες ταινιών χρησιμοποιώντας ξένα κλειδιά.
 - film_id
 - category_id

Lab schema: *Film*

FILM		
FILM_ID	NUMBER(22)	PK
TITLE	VARCHAR2(255)	
DESCRIPTION	CLOB	N
RELEASE_YEAR	VARCHAR2(4)	N
LANGUAGE_ID	NUMBER(22)	FK
ORIGINAL_LANGUAGE_ID	NUMBER(22)	N FK
RENTAL_DURATION	NUMBER(22)	
RENTAL_RATE	NUMBER(4,2)	
LENGTH	NUMBER(22)	N
REPLACEMENT_COST	NUMBER(5,2)	
RATING	VARCHAR2(10)	N
SPECIAL_FEATURES	VARCHAR2(100)	N
LAST_UPDATE	TIMESTAMP	

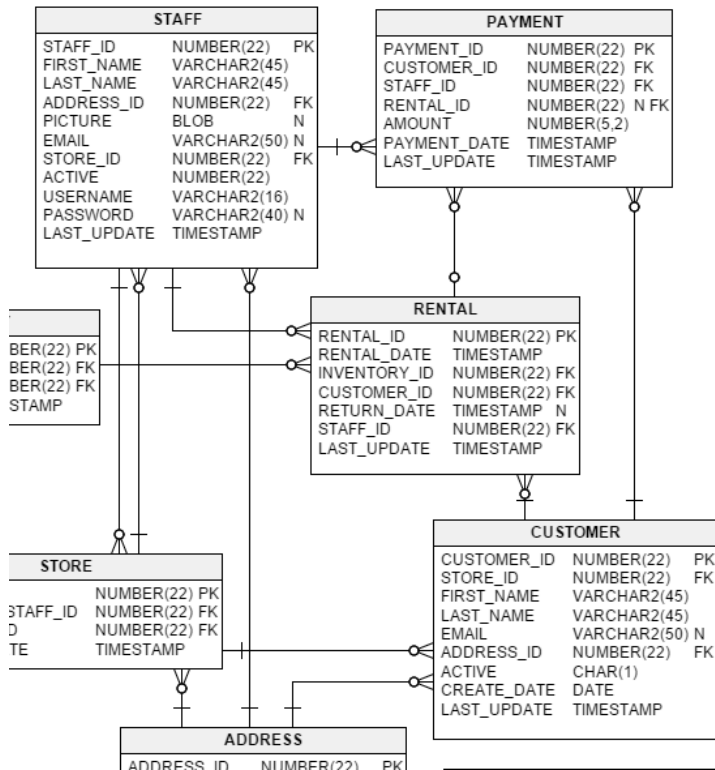
- Ο πίνακας **film** περιέχει όλες τις ταινίες που έχουν τα καταστήματα. Τα πραγματικά αντίγραφα βρίσκονται στο πίνακα inventory.
 - film_id (πρωτεύον κλειδί)
 - title
 - description
 - release_year
 - language_id
 - original_language_id
 - rental_duration
 - rental_rate
 - length
 - replacement_cost
 - rating (G, PG, PG-13, R, ή NC-17)
 - special_features (Trailers, Commentaries, Deleted Scenes, Behind the Scenes)

Lab schema: *Inventory, Rental*



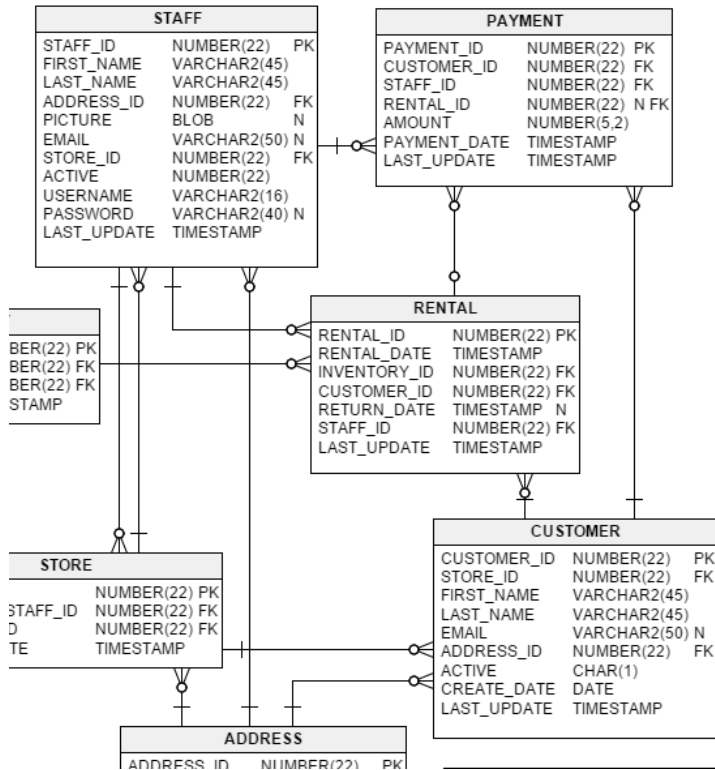
- Ο πίνακας **inventory** περιέχει κάθε αντίγραφο μιας ταινίας σε κάθε κατάσταση. Χρησιμοποιεί ξένα κλειδιά και σχετίζεται με τον πίνακα rental και τον πίνακα store.
 - inventory_id (πρωτεύον κλειδί)
 - film_id
 - store_id
- Ο πίνακας **rental** περιέχει κάθε ενοικίαση που έχει γίνει για κάθε στοιχείο του πίνακα inventory και δείχνει τους πίνακες inventory, customer και staff.
 - rental_id (πρωτεύον κλειδί)
 - rental_date
 - inventory_id
 - customer_id
 - return_date

Lab schema: *Customer*



- Ο πίνακας **customer** περιέχει τους πελάτες και αναφέρεται στους πίνακες rental και payment και ενώνεται με τους πίνακες address και store χρησιμοποιώντας ένα ξένο κλειδί.
 - customer_id (πρωτεύον κλειδί)
 - store_id
 - first_name
 - last_name
 - address_id
 - active

Lab schema: *Payment*



- Ο πίνακας **payment** περιέχει κάθε πληρωμή που έχει γίνει από ένα πελάτη και δείχνει τους πίνακες customer, staff και rental.
 - payment_id (πρωτεύον κλειδί)
 - customer_id
 - staff_id
 - rental_id
 - amount
 - payment_date

Lab 1 Intro: Wrap Up

1. [X] Διαδικαστικά
2. [X] DBMS Installation
3. [X] Lab schema Installation
4. [X] Lab schema Explanation

Lab 1 Intro: Wrap Up

Next time make sure 

Installation	Status
DBMS	
DBMS Clients	
Web Dev Stack	
Sample lab schema	

Απορίες <https://discord.gg/9UvTeWNJzs> 

